

Lab 14:

- Firebase Authentication Session Handling
- Auto Login after restart
- Logout
- Firebase Realtime Database
- Add Post Screen
- Floating Action Button
- Loading Indicator
- Validation

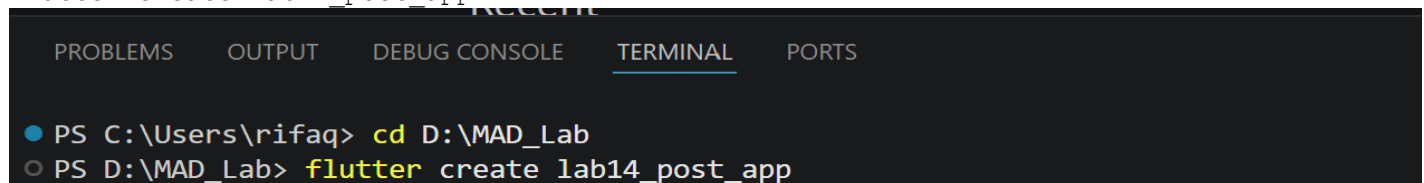
We already did Lab 12 & 13, so this lab will continue from them.

STEP 1 : Create Project

terminal:

```
cd D:\MAD_Lab
```

```
flutter create lab14_post_app
```



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

● PS C:\Users\rifaq> cd D:\MAD_Lab
○ PS D:\MAD_Lab> flutter create lab14_post_app
```

Then:

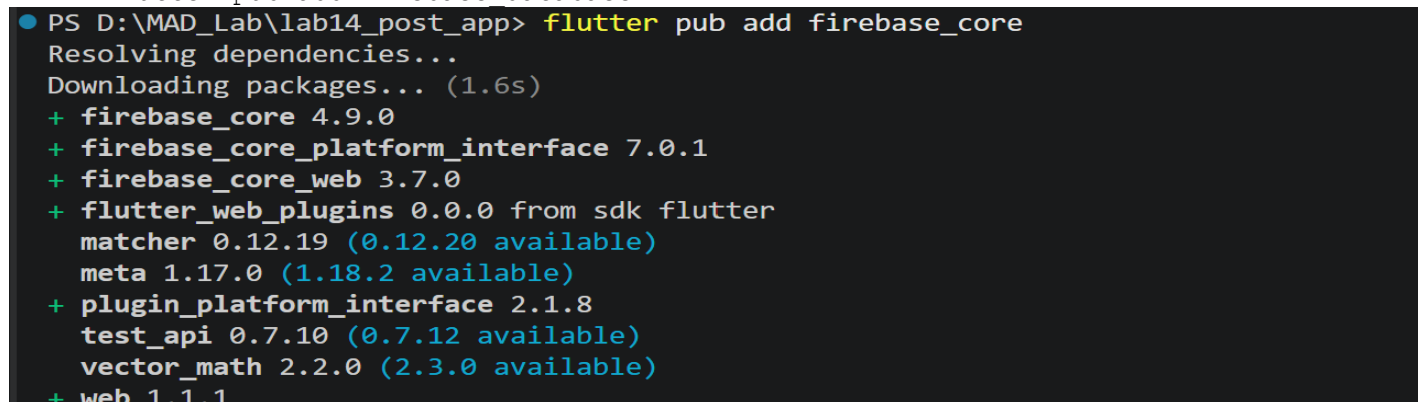
```
cd lab14_post_app
```



```
● PS D:\MAD_Lab> cd lab14_post_app
○ PS D:\MAD_Lab\lab14_post_app> 
```

STEP 2 : Add Firebase Packages

- flutter pub add firebase_core
- flutter pub add firebase_auth
- flutter pub add firebase database



```
● PS D:\MAD_Lab\lab14_post_app> flutter pub add firebase_core
Resolving dependencies...
Downloading packages... (1.6s)
+ firebase_core 4.9.0
+ firebase_core_platform_interface 7.0.1
+ firebase_core_web 3.7.0
+ flutter_web_plugins 0.0.0 from sdk flutter
+ matcher 0.12.19 (0.12.20 available)
+ meta 1.17.0 (1.18.2 available)
+ plugin_platform_interface 2.1.8
+ test_api 0.7.10 (0.7.12 available)
+ vector_math 2.2.0 (2.3.0 available)
+ web 1.1.1
```

```
PS D:\MAD_Lab\lab14_post_app> flutter pub add firebase_auth
Resolving dependencies...
Downloading packages... (1.7s)
+ _flutterfire_internals 1.3.71
+ firebase_auth 6.5.1
+ firebase_auth_platform_interface 9.0.1
+ firebase_auth_web 6.2.1
+ http 1.6.0
+ http_parser 4.1.2
  matcher 0.12.19 (0.12.20 available)
  meta 1.17.0 (1.18.2 available)
  test_api 0.7.10 (0.7.12 available)
+ typed_data 1.4.0
  vector_math 2.2.0 (2.3.0 available)
Changed 7 dependencies!
```

```
PS D:\MAD_Lab\lab14_post_app> flutter pub add firebase_database
Resolving dependencies...
Downloading packages... (1.4s)
+ firebase_database 12.4.1
+ firebase_database_platform_interface 0.4.0+1
+ firebase_database_web 0.2.7+8
  matcher 0.12.19 (0.12.20 available)
  meta 1.17.0 (1.18.2 available)
  test_api 0.7.10 (0.7.12 available)
  vector_math 2.2.0 (2.3.0 available)
Changed 3 dependencies!
4 packages have newer versions incompatible with dependency constraints.
Try `flutter pub outdated` for more information.
PS D:\MAD_Lab\lab14_post_app>
```

STEP 3 : Create Firebase Project

[Firebase Console](#)

Create new project:

lab14firebase

Disable analytics.



STEP 4 : Add Android App

Inside Firebase project:

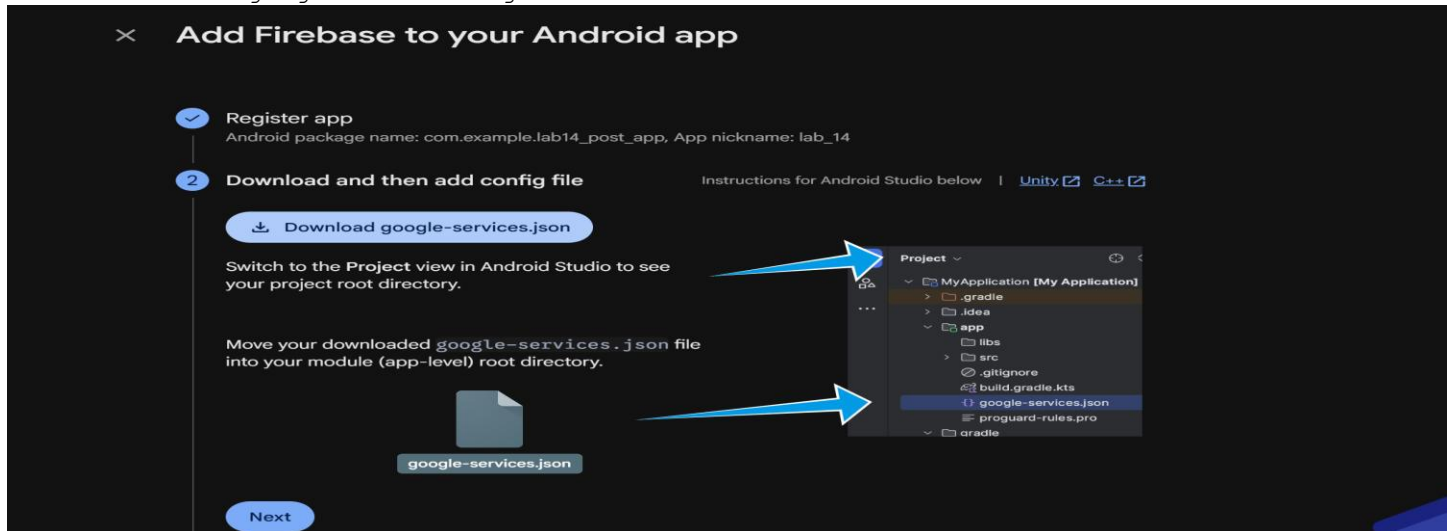
- Click Android

Package name:

- com.example.lab14_post_app

Then:

- Register App
- Download google-services.json



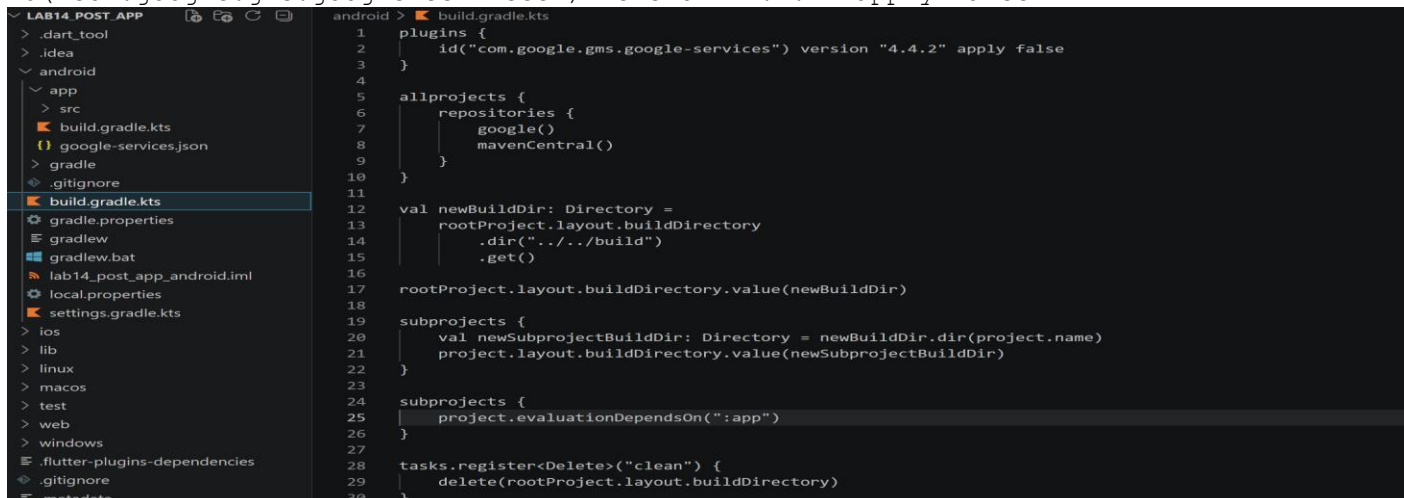
- Put it inside:
- android/app

STEP 5 : Configure Gradle

android/build.gradle.kts

Inside plugins add:

`id("com.google.gms.google-services") version "4.4.2" apply false`

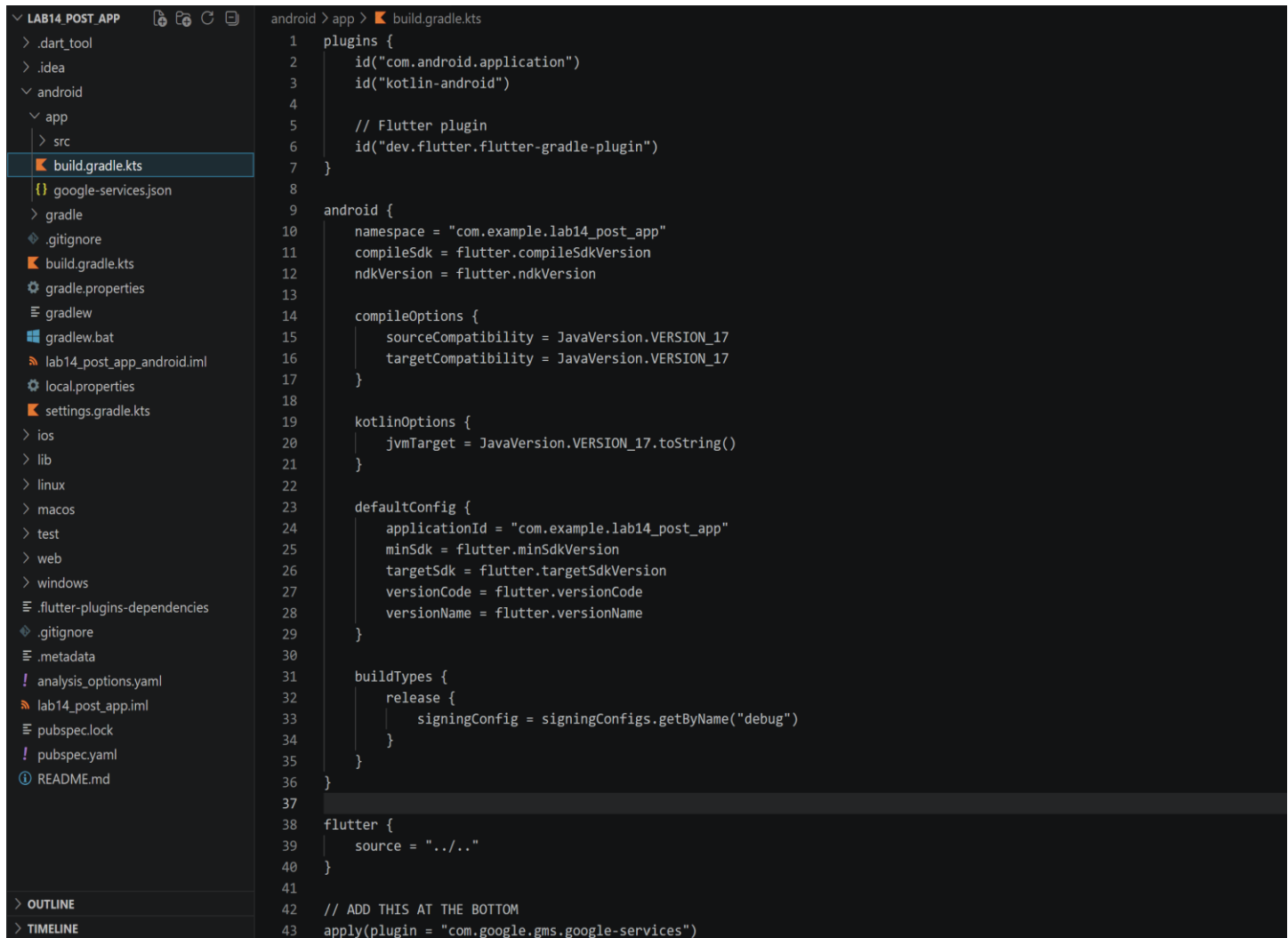


Open:

android/app/build.gradle.kts

Add at bottom:

```
apply(plugin = "com.google.gms.google-services")
```

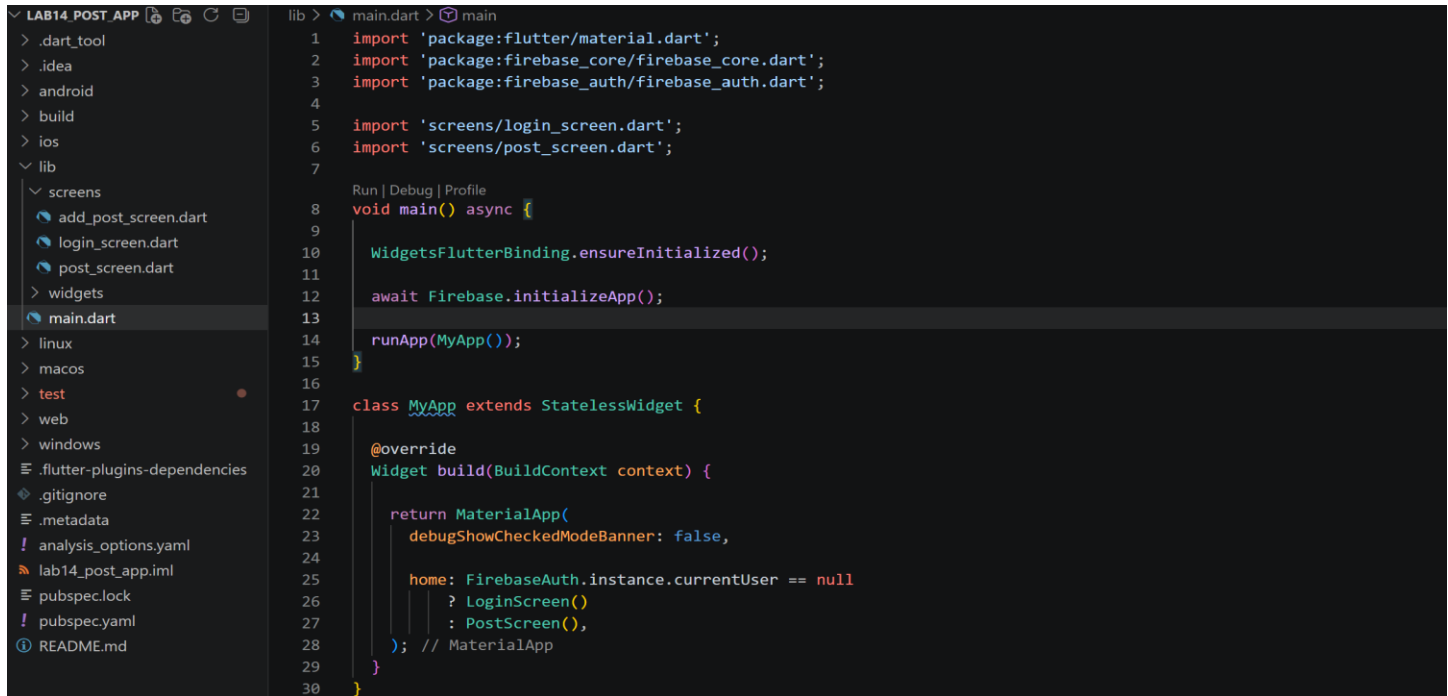


```
1  plugins {
2      id("com.android.application")
3      id("kotlin-android")
4
5      // Flutter plugin
6      id("dev.flutter.flutter-gradle-plugin")
7  }
8
9  android {
10     namespace = "com.example.lab14_post_app"
11     compileSdk = flutter.compileSdkVersion
12     ndkVersion = flutter.ndkVersion
13
14     compileOptions {
15         sourceCompatibility = JavaVersion.VERSION_17
16         targetCompatibility = JavaVersion.VERSION_17
17     }
18
19     kotlinOptions {
20         jvmTarget = JavaVersion.VERSION_17.toString()
21     }
22
23     defaultConfig {
24         applicationId = "com.example.lab14_post_app"
25         minSdk = flutter.minSdkVersion
26         targetSdk = flutter.targetSdkVersion
27         versionCode = flutter.versionCode
28         versionName = flutter.versionName
29     }
30
31     buildTypes {
32         release {
33             signingConfig = signingConfigs.getByName("debug")
34         }
35     }
36 }
37
38 flutter {
39     source = "../.."
40 }
41
42 // ADD THIS AT THE BOTTOM
43 apply(plugin = "com.google.gms.google-services")
```

STEP 6 : Initialize Firebase

Open:

lib/main.dart



```
1 import 'package:flutter/material.dart';
2 import 'package:firebase_core/firebase_core.dart';
3 import 'package:firebase_auth/firebase_auth.dart';
4
5 import 'screens/login_screen.dart';
6 import 'screens/post_screen.dart';
7
8 void main() async {
9   WidgetsFlutterBinding.ensureInitialized();
10  await Firebase.initializeApp();
11  runApp(MyApp());
12 }
13
14 class MyApp extends StatelessWidget {
15   @override
16   Widget build(BuildContext context) {
17     return MaterialApp(
18       debugShowCheckedModeBanner: false,
19       home: FirebaseAuth.instance.currentUser == null
20         ? LoginScreen()
21         : PostScreen(),
22     ); // MaterialApp
23   }
24 }
```

STEP 7 : Create Folders

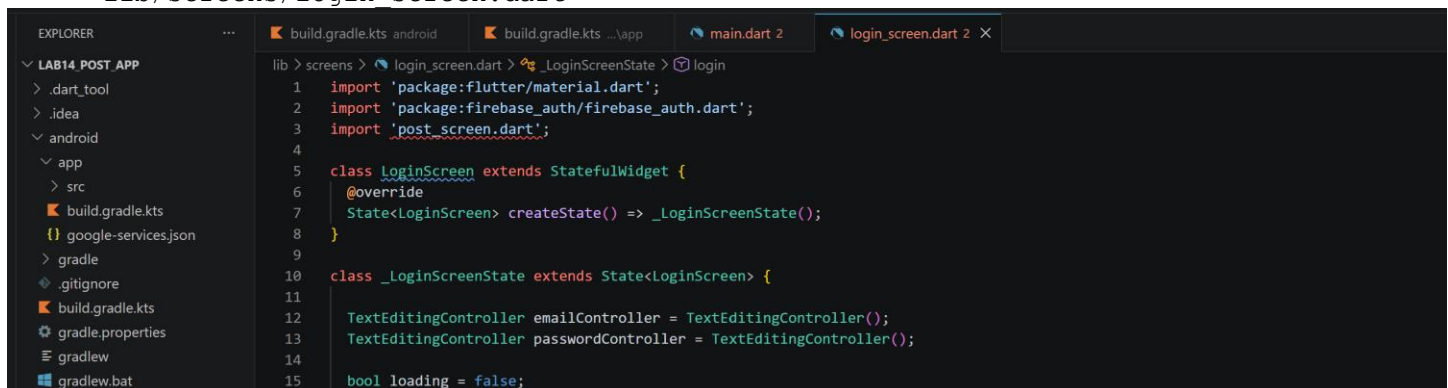
Inside lib

Create:

- screens
- widgets

STEP 8 : Create Login Screen

- Create file:
- lib/screens/login_screen.dart



```
1 import 'package:flutter/material.dart';
2 import 'package:firebase_auth/firebase_auth.dart';
3 import 'screens/post_screen.dart';
4
5 class LoginScreen extends StatefulWidget {
6   @override
7   State<LoginScreen> createState() => _LoginScreenState();
8 }
9
10 class _LoginScreenState extends State<LoginScreen> {
11   TextEditingController emailController = TextEditingController();
12   TextEditingController passwordController = TextEditingController();
13   bool loading = false;
14 }
```

```

16 login() async {
17
18   setState(() {
19     loading = true;
20   });
21
22   try {
23
24     await FirebaseAuth.instance.signInWithEmailAndPassword(
25       email: emailController.text,
26       password: passwordController.text,
27     );
28
29     Navigator.pushReplacement(
30       context,
31       MaterialPageRoute(builder: (_) => PostScreen()),
32     );
33   } catch (e) {
34
35     ScaffoldMessenger.of(context).showSnackBar(
36       SnackBar(content: Text(e.toString())),
37     );
38   }
39
40   setState(() {
41     loading = false;
42   });
43
44   @override
45   Widget build(BuildContext context) {
46
47     return Scaffold(
48
49       appBar: AppBar(title: Text("Login")),
50
51       body: Padding(
52         padding: EdgeInsets.all(20),
53
54         child: Column(
55
56           children: [
57
58             TextField(
59               controller: emailController,
60               decoration: InputDecoration(
61                 hintText: "Email",
62               ), // InputDecoration
63             ), // TextField
64
65             SizedBox(height: 20),
66
67             TextField(
68               controller: passwordController,
69               decoration: InputDecoration(
70                 hintText: "Password",
71               ), // InputDecoration
72             ), // TextField
73
74             SizedBox(height: 30),
75
76             loading
77               ? CircularProgressIndicator()
78               : ElevatedButton(
79                 onPressed: login,
80                 child: Text("Login"),
81               ), // ElevatedButton
82
83           ],
84         ), // Column
85       ), // Padding
86     ); // Scaffold
87   }
88 }
89
90
91
92

```

STEP 9 : Create Post Screen

Create:

lib/screens/post_screen.dart

```
lib > screens > post_screen.dart > PostScreen > build
1  import 'package:flutter/material.dart';
2  import 'package:firebase_auth/firebase_auth.dart';
3
4  import 'add_post_screen.dart';
5  import 'login_screen.dart';
6
7  class PostScreen extends StatelessWidget {
8
9    logout(BuildContext context) async {
10
11      await FirebaseAuth.instance.signOut();
12
13      Navigator.pushReplacement(
14        context,
15        MaterialPageRoute(builder: (_) => LoginScreen()),
16      );
17    }
18
19    @override
20    Widget build(BuildContext context) {
21
22      return Scaffold(
23
24        appBar: AppBar(
25          title: Text("PostScreen"),
26
27          actions: [
28
29            IconButton(
30              onPressed: () {
31                logout(context);
32              },
33              icon: Icon(Icons.logout),
34            ) // IconButton
35          ],
36        ), // AppBar
37
38        floatingActionButton: FloatingActionButton(
39
40          child: Icon(Icons.add),
41
42          onPressed: () {
43            Navigator.push(
44              context,
45              MaterialPageRoute(
46                builder: (_) => AddPostScreen(),
47              ), // MaterialPageRoute
48            );
49          },
50        ), // FloatingActionButton
51      ); // Scaffold
52    }
53  }
54
55 }
```


STEP 10 : Create AddPostScreen

lib/screens/add_post_screen.dart

```

1  import 'package:firebase_database/firebase_database.dart';
2  import 'package:flutter/material.dart';
3
4  class AddPostScreen extends StatefulWidget {
5    @override
6    State<AddPostScreen> createState() => _AddPostScreenState();
7  }
8
9  class _AddPostScreenState extends State<AddPostScreen> {
10
11    TextEditingController postController = TextEditingController();
12
13    bool loading = false;
14
15    final databaseRef = FirebaseDatabase.instance.ref("Posts");
16
17    addPost() async {
18
19      if(postController.text.isEmpty){
20
21        ScaffoldMessenger.of(context).showSnackBar(
22          SnackBar(content: Text("Enter something")),
23        );
24
25        return;
26      }
27
28      setState(() {
29        loading = true;
30      });
31
32      String id = DateTime.now().millisecondsSinceEpoch.toString();
33
34      await databaseRef.child(id).set({
35
36        "id": id,
37        "title": postController.text,
38      });
39
40      setState(() {
41        loading = false;
42      });
43
44      postController.clear();
45      Navigator.pop(context);
46    }
47
48    @override
49    Widget build(BuildContext context) {
50
51      return Scaffold(
52
53        appBar: AppBar(
54          title: Text("AddPostScreen"),
55        ), // AppBar
56
57        body: Padding(
58
59          padding: EdgeInsets.all(20),
60
61          child: Column(
62
63            children: [
64
65              TextField(
66
67                controller: postController,
68
69                maxLines: 5,
70
71                decoration: InputDecoration(
72                  hintText: "Write something...",
73                  border: OutlineInputBorder(),
74                ), // InputDecoration
75              ), // TextField
76
77              SizedBox(height: 30),
78
79              loading
80                ? CircularProgressIndicator()
81                : ElevatedButton(
82                  child: Text("Add"),
83                ), // ElevatedButton
84
85            ],
86          ), // Column
87        ), // Padding
88      ); // Scaffold
89
90    }
91  }
92
93

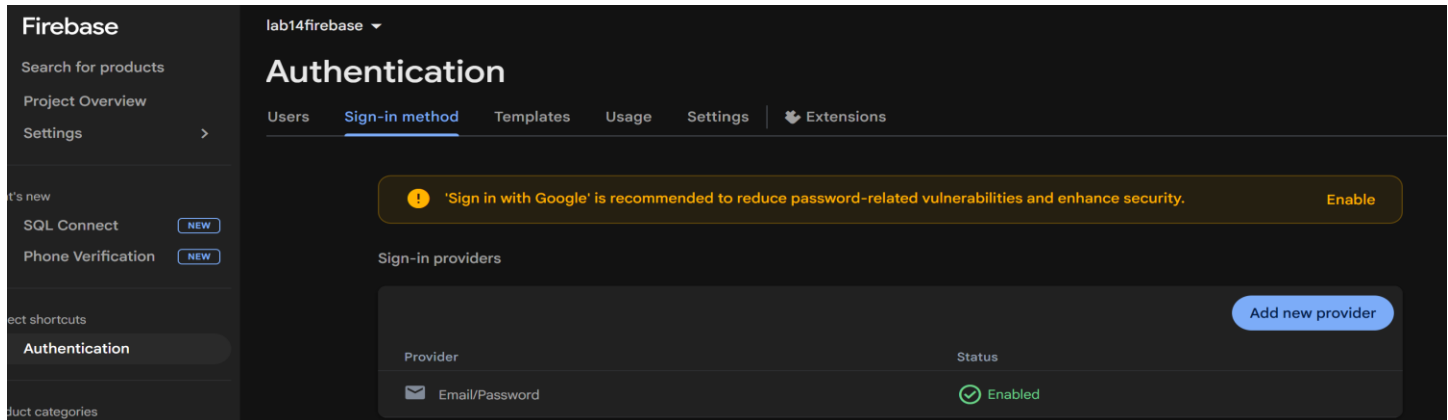
```


STEP 11 : Enable Email Authentication

Firebase Console → Authentication → Get Started

Enable:

Email/Password



STEP 12 : Create User

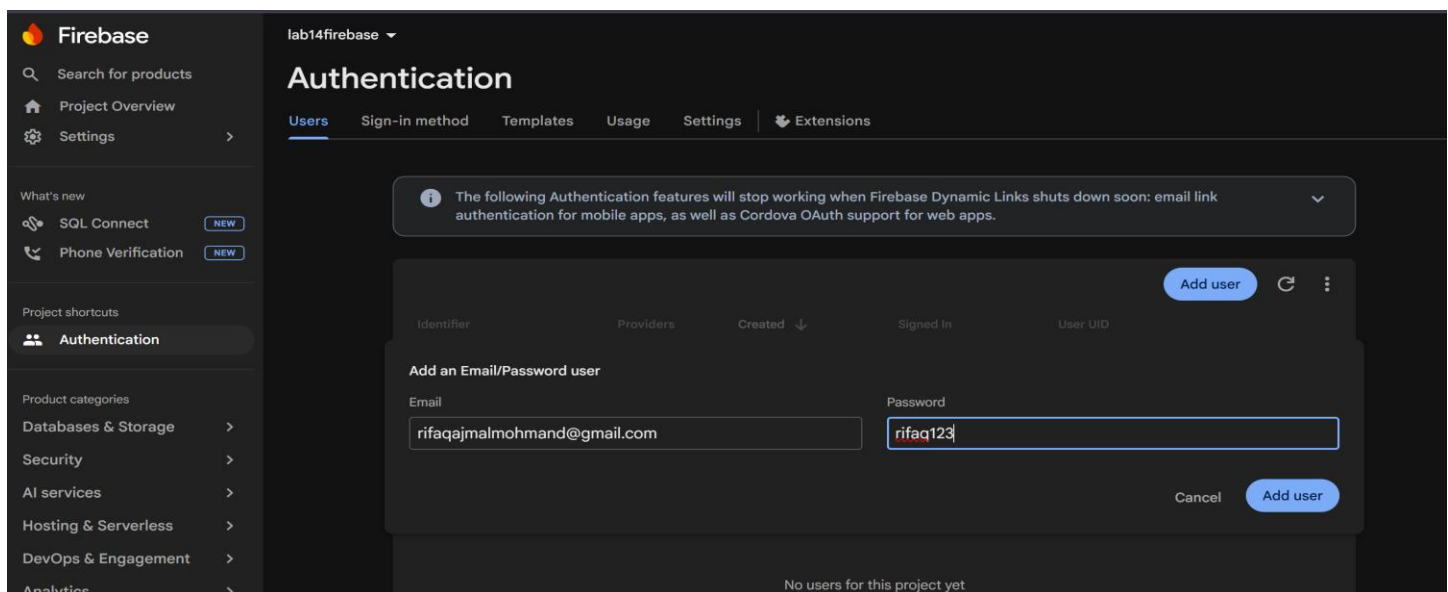
Inside Firebase Authentication:

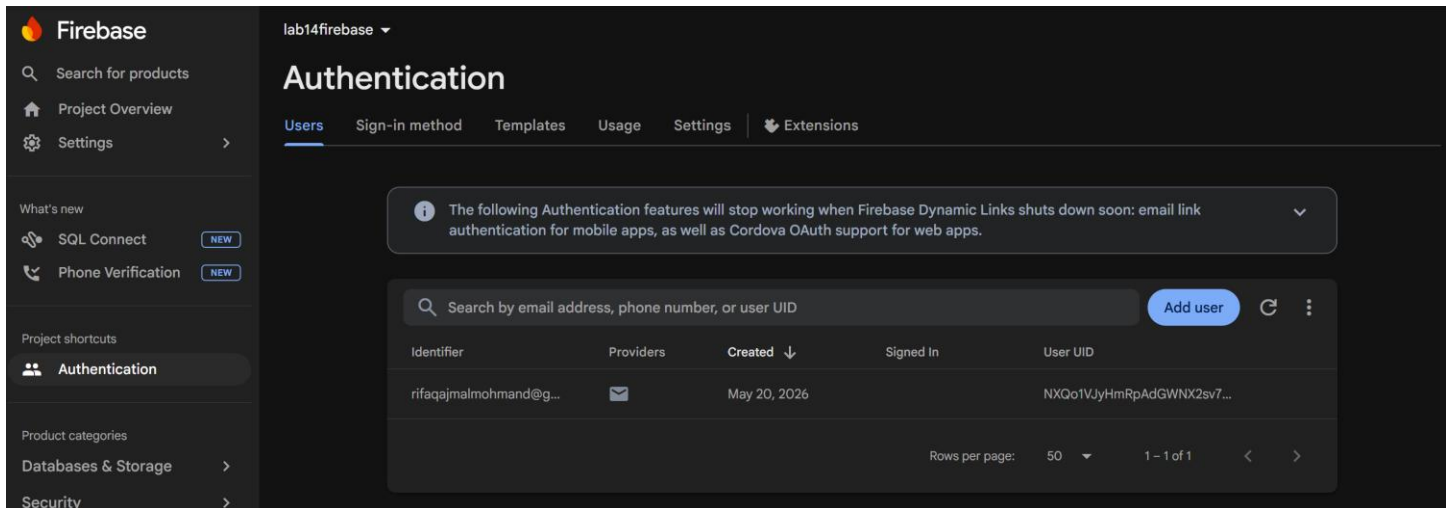
Add User

Example:

Email: test@gmail.com

Password: 123456

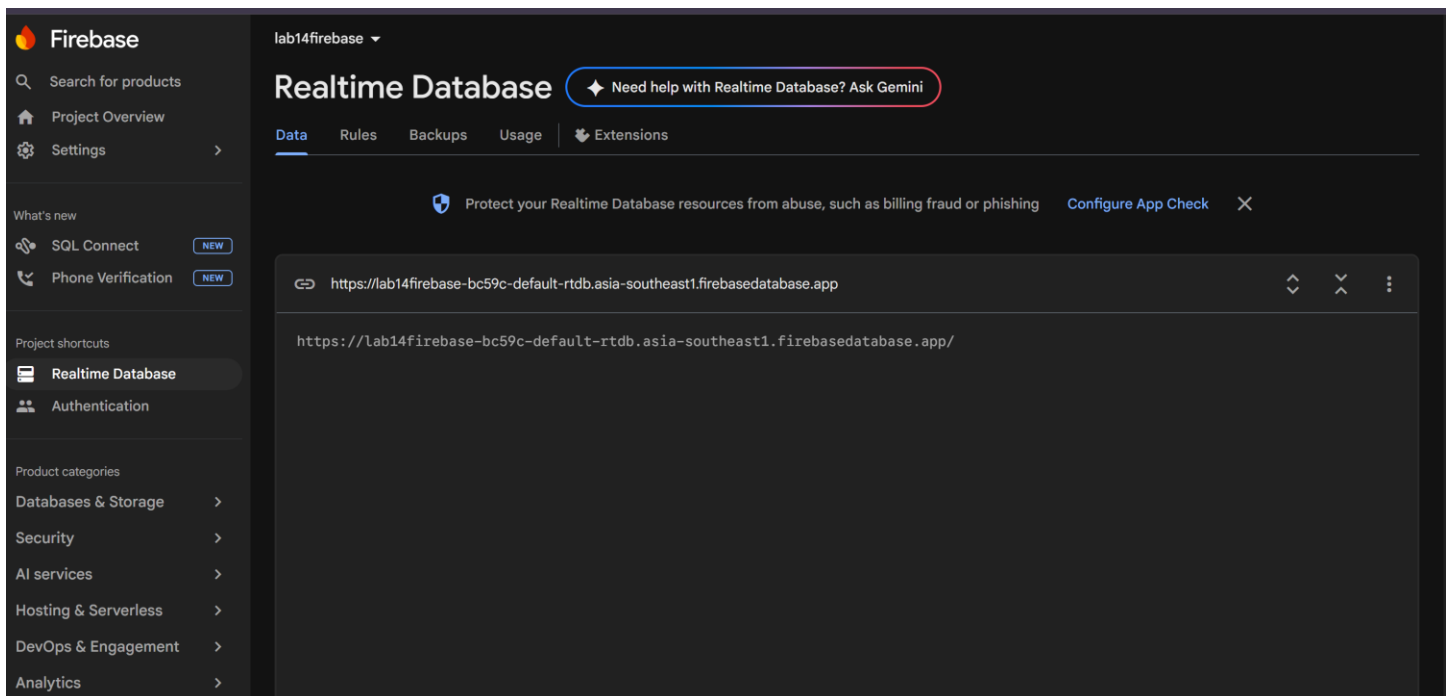




STEP 13 : Enable Realtime Database

Firebase → Build → Realtime Database

- Create Database
- Choose:
Start in test mode



STEP 13.1 : SET DATABASE RULES

- Click:
 - Rules

(tab at top beside Data)

- Replace everything with:
- Publish



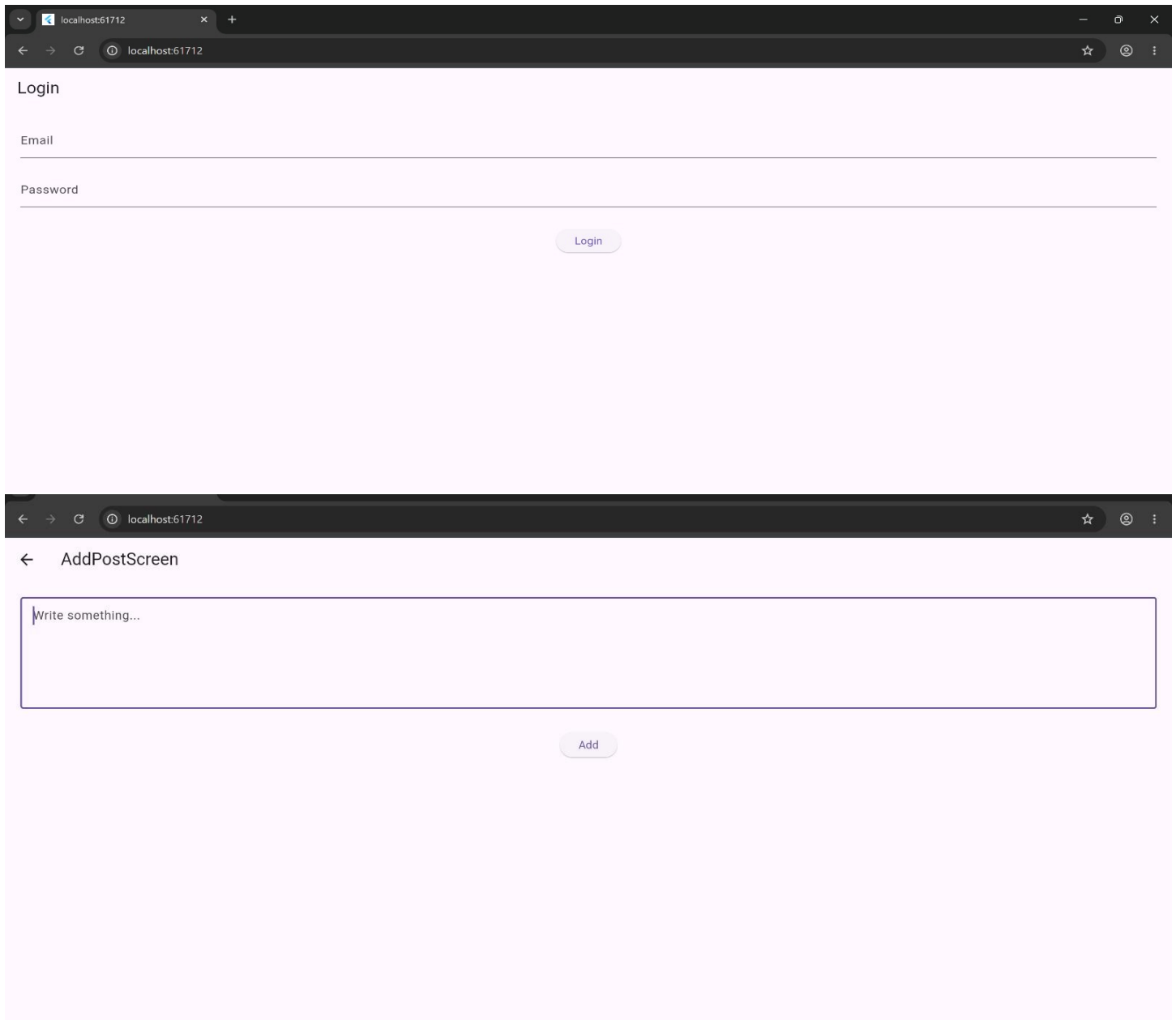
Your security rules are defined as public, so anyone can steal, modify, or delete data in your database

```
1  {
2    "rules": {
3      ".read": true,
4      ".write": true
5    }
6  }
```

STEP 14 : RUN THE APP

Run:

```
flutter devices  
flutter run
```



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BSCS – 6th semester

MAD - Lab

Lab #14

